Econ 402: Econometrics Name: _____

Exam 1

Some useful formulae:

$$Var(X) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1},\ SD(X) = \sqrt{Var(X)},\ \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

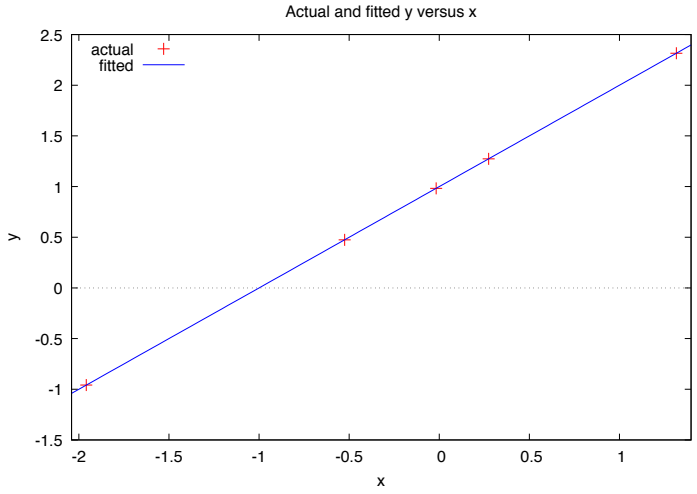
$$b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})} = \frac{Cov(X,Y)}{Var(X)} = Corr(X,Y) \cdot \frac{SD(Y)}{SD(X)}$$

$$a = \bar{y} - b\bar{x}$$

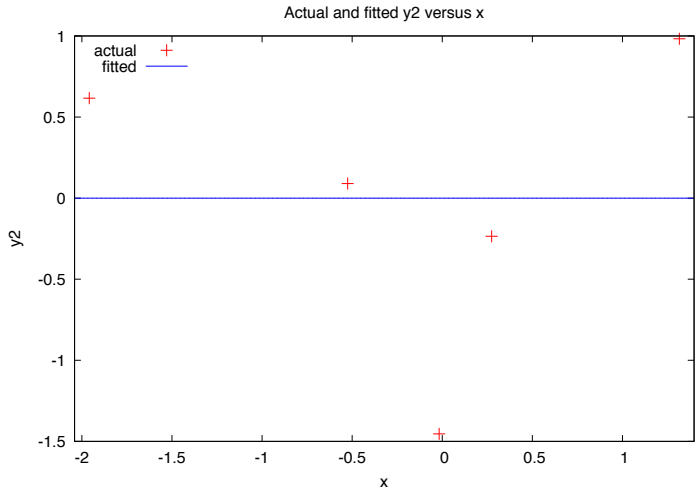
$$R^2 = \frac{b^2 \sum_{i=1}^{n}(x_i - \bar{x})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^{n} e_i^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

1. The figures below show the actual and predicted values of y for two different regressions (the solid lines, labelled "fitted," represent the predicted values/regression lines). Determine the R^2 for each regression, and provide a (mathematical) justification for your response. *Hint: you do not need a calculator to answer this question.*

   (a) Regression 1:



(b) Regression 2:



2. Define (as precisely as you can) the following assumptions that we made about the disturbance term ε and provide an intuitive one-sentence interpretation of each assumption:

   (a) Homoskedasticity

   (b) No autocorrelation

   (c) Mean-zero errors

3. In class, we saw that the OLS estimate b is both an unbiased and a consistent estimate of the population slope coefficient β. What is the difference between unbiasedness and consistency?

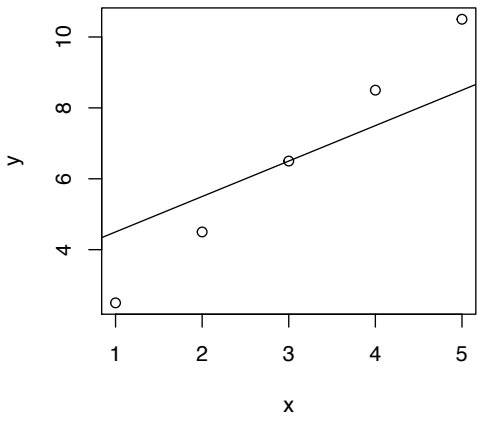
4. State the Gauss-Markov theorem and briefly explain why we care about it.

5. In class, we derived the following properties of the residuals, e_i:

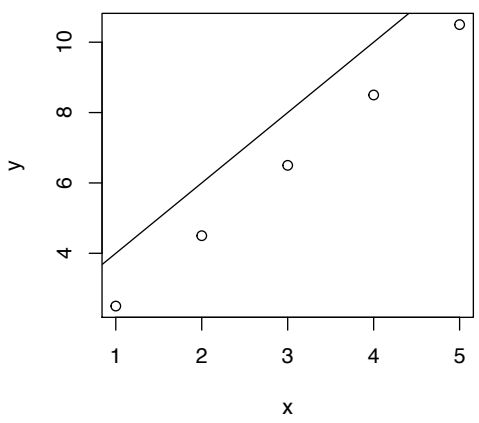
$$\bar{e} = 0 \qquad \text{(Property 1)}$$
$$Cov(e, x) = 0 \qquad \text{(Property 2)}$$

Below are two hypothetical regression lines, each of which violates one of the above properties (hence they are not truly least-squares regression lines). State which property each line violates, justifying your answer.

(a)



(b)



6. The following gretl output provides summary statistics on hours of work (HOURS, measured in hundreds of hours worked) and marginal tax rates (MTR, the rate at which an additional dollar of income is taxed) in 1987 for a sample of 753 women.

	Mean	Median	S.D.	Min	Max
MTR	0.6789	0.6915	0.08350	0.4415	0.9415
HOURS	7.406	2.880	8.713	0.000	49.50

Gretl also reports that the correlation between HOURS and MTR is

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corr(HOURS, MTR) = -0.19192674
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(a) What is the equation for a regression of HOURS on MTR (recall that this means that HOURS is the y variable and MTR is the x variable)? *Your answer should include both the constant term a and the slope coefficient b.*

(b) How do you interpret the slope coefficient b from this regression? *Be precise.* What do you conclude about the effect of taxes on labor supply?

(c) Gretl reports that the sum of squared errors is 54987.98. Show that the R^2 from the regression is about 0.037.

(d) What does it mean that the R^2 is 0.037? *Be as specific as you can.*